QNX on Raspberry Pi

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QNX Everywhere on Raspberry Pi

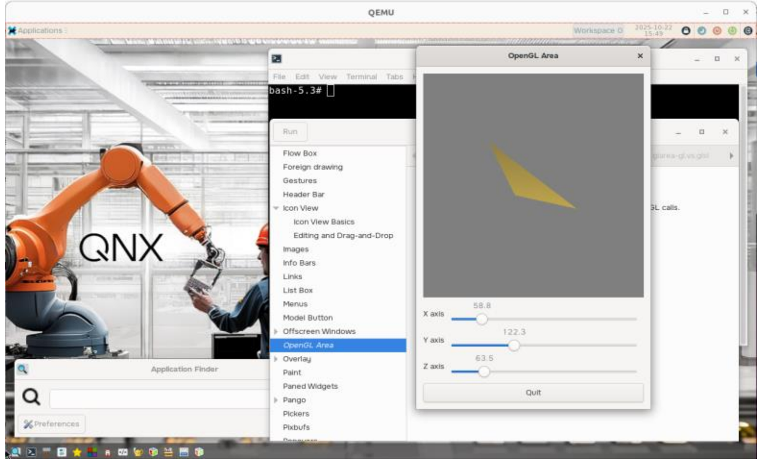
Quick Start Target Image:

- no playing with BSPs, build scripts, configurations, policies, etc
- works out of the box and includes some basic samples

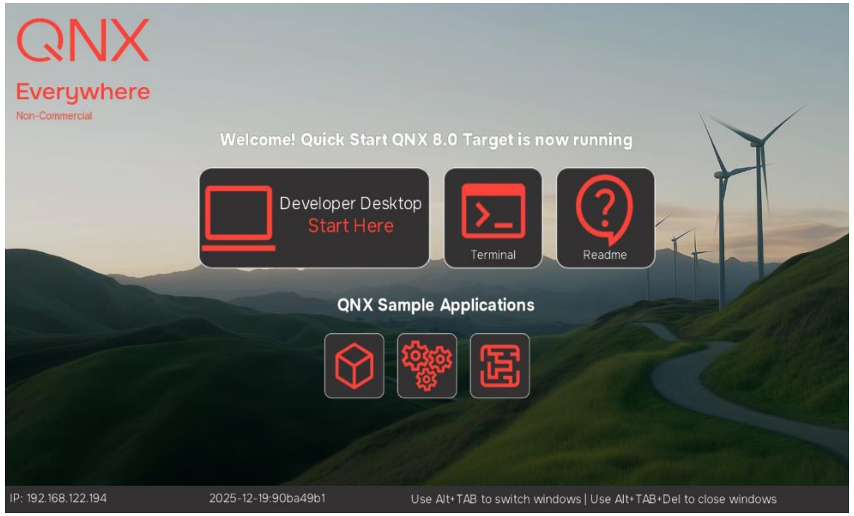
Custom Target Image:

- uses the BSP, mkqnximage, snippets, scripts
- builds a fully custom image with exactly what you want in it

Released recently: QNX self-hosted desktop!



We’ve also open-sourced our BSPs, for fully customized builds



Web search:

"QNX Everywhere Documentation"

www.qnx.com/developers/docs/qnxeverywhere/index.html

Next up: Pi distribution...

Your Raspberry Pi 5 Kit!

Kit includes:

- Raspberry Pi 4 or 5 and accessories (case, power, heatsink, fans)
- Micro-SD card with image of QNX 8.0 for Pi (already loaded in the Pi)
- Your Micro-SD card is specific to you! Your micro-SD card came with a hostname – note it!
- A Logitech C270 HD Webcam

Here's the plan:

- Download a VNC Client (https://sourceforge.net/projects/tigervnc/files/stable/1.16.2/)
  - Google "Tiger VNC", download, install
- Distribute Pis
- Power them up, try to connect with VNC (<your-hostname.local>:5900 for VNC)
- Clone the repositories and setup camera
- Try the sample to get the camera working with a Hand Landmarking Model

Next up: Pi set up...

QNX Project on QDD

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AI Hand Landmark Demo - Step-by-Step

Software repository: https://github.com/qnx/aiworkshop

1. Connect to your Pi over VNC (VNC must be used to see the application
    1. Click the icon for the Desktop
    2. In the Applications menu, open Terminal
2. Clone the repository we need:

```
git clone https://github.com/qnx/aiworkshop
```

3. Follow the Hand Landmarking README.md to connect your camera
4. Install all dependencies and ML model

```
$ cd aiworkshop && ./configure.sh
```

5. Build the demo

```
$ ./build.sh
```

6. Try the code!

```
$ ./start.sh --ros-args -p camera_unit:=5
```

How to Connect to Your Raspberry Pi

Connect to the network on your laptop: use SSID "QNX", password "everywhere”.

Connect to your Pi & try some commands:

- $ ping <your-hostname>.local
- $ ssh qnxuser@<your-hostname>.local
  OR
  $ ssh qnxuser@10.20.30.40
- **VNC Client:** <your-hostname>.local:5900

Default username and password is qnxuser/qnxuser.

The tools sudo and su are available (root/root).

Common QNX Commands

\# pidin - List processes

\# slog2info - Output logs (must be root!)

\# slay <p>

Send a signal to p - (By default, SIGTERM)

\# use <app> - Get usage info on app

\# top - Get system utilization

\# hogs - Print the system utilization

Using Windows? To SSH use the "Terminal" app and:

> ssh **-m hmac-sha2-256** qnxuser@qnxpi00.local

For file transfers:

> scp **-o MACs=hmac-sha2-256** <file> qnxuser@<your-ip>:~

The software repository: https://github.com/qnx/aiworkshop

FOLLOW THE README!

AI Hand Landmark Demo

Stretch goal

- Update the code in `handtrack_gui.hpp` to pick a display name and colour for hand landmarking. Make sure you rebuild the application and restart to see the changes.
- Change confidence values to see how it affects the detection see README for all options.
- Implement basic gesture recognition by using the position of the landmarks (thumbs up, thumbs down, hand open, hand closed)

Possible new features

- Update the UI to support dynamically changing ML configuration settings such has confidence values during runtime with GTK4 widgets.
- Add Pose Landmarking using MediaPipe pose_landmarker
- Add ML Gesture recognition as a next step on top of the existing models.
- Convert this project to use MediaPipe directly using our port https://github.com/qnx-ports/mediapipe/.
- Update the demo so two players can play pong against each other with the hand position as the paddles.

Important Links

Download QNX Everywhere: https://qnx.com/getqnx

QNX for VS Code: https://www.qnx.com/developers/docs/8.0/com.qnx.doc.qnxtoolkit.user_guide/topic/quickstart_guide.html

Using QNX on a Mac: https://devblog.qnx.com/how-to-use-macos-to-develop-for-qnx-8-0/

QNX Open-source Ports: https://oss.qnx.com/

QNX GitLab: https://gitlab.com/qnx/

QNX GitHub: https://github.com/qnx/

QNX Free Online Training: https://learning.qnx.com/

QNX Discord Server: https://discord.gg/tQVpvCzD

QNX Subreddit: https://www.reddit.com/r/QNX/

Thank you